



GENERATIVE GROOVEBOX
DESIGNED BY VADIM MINKIN

CONTENTS

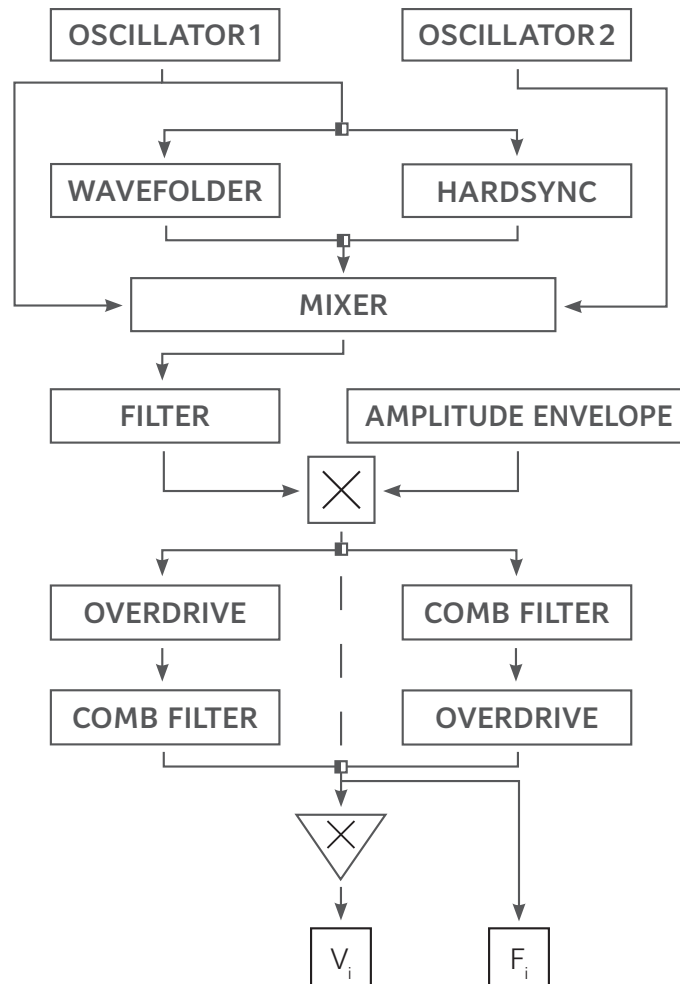
INTRODUCTION	4
PART 1. USER INTERFACE	6
APPEARANCE	6
CONNECTION	6
USB	6
MIDI	7
Audio	7
CONTROL	7
Display	8
Incremental encoders with a built-in button	10
Track selector	10
Page selector	10
Interacting with the interface	10
Summary of all options for interaction	
between the track selector and the encoder	12
Page selection	13
PART 2. PARAMETERS AND INTERNAL STRUCTURE	16
SUMMARY TABLE OF THE LOCATION	
OF PARAMETERS ON THE PAGES	18
WAVEFORM GROUP	19
VOLUME ENVELOPE GROUP	29
MODULATION ENVELOPE GROUP	36
GROUP XMOD1	39
XMOD2 GROUP	41
CFG GROUP	43
PART 3. GENERATIVE SEQUENCER	47
ENVELOPE FEATURES	47
Trigger Exchange	47
Pattern Mask	48
Rhythmic Nature	48
Modulator Quantization	49
Note quantization	50
Combining with an external sequencer or keyboard	51
Examples	51

MISCELLANEOUS

MANAGING PRESETS	52
Saving and Loading	52
Exporting and Importing Presets.	53
FIRMWARE UPDATE	54
VALUE SCALES	54
POLYPHONY	55
Polyphony and MIDI channels	55
Track selector modes	56
Activation and Hold indicator	
Hold, Drone and Edit	56
Effect of MIDI Notes	57
MIDI PARAMETER TABLE	58
CREDITS.	61

INTRODUCTION

KALEIDOSCOPE is a compact generative synthesizer and digital laboratory for sound exploration.



Architecture of the KALEIDOSCOPE Sound Path.

KALEIDOSCOPE is 4 digital synthesizers with the same architecture in one case.

The architecture is so flexible that it allows working with different types of synthesis:

- subtractive,
- additive,
- fm,
- physical modeling.

The power of the instrument is not in the number of modules or their internal complexity. KALEIDOSCOPE has a very flexible and convenient control system that feels at your fingertips.

In KALEIDOSCOPE, you can place different timbres on the tracks. But these timbres are able to work together as parts of the overall sound. The final sound is not just the sum of the sounds of individual tracks, but a real superposition — that is, the result of the simultaneous influence of tracks on each other.

The emphasis is on interaction — tracks can modulate each other in many interesting combinations. Parametric cross-modulation of most parameters, side chains, as well as advanced FM synthesis, which has no analogues on the market.

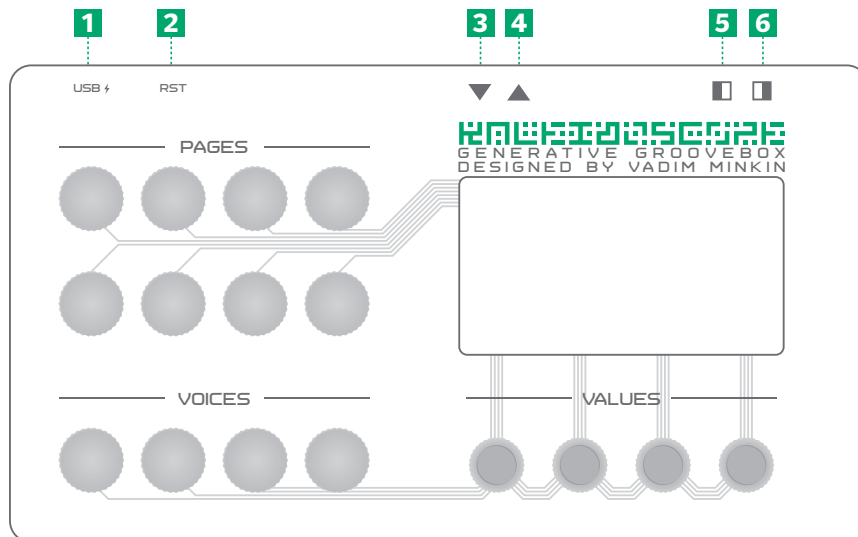
KALEIDOSCOPE gives the user:

- a powerful, but not overloaded with details set of modules,
- the ability of voices to modulate most of each other's parameters,
- a uniform and intuitive control interface,
- a compact form factor that allows you to use the instrument anywhere.

PART 1. USER INTERFACE

APPEARANCE

The figure shows a KALEIDOSCOPE with numbered interfaces.



- 1 USB
- 2 Reset
- 3 MIDI-Out
- 4 MIDI-Out
- 5 Audio Out 1-2
- 6 Audio Out 3-4

CONNECTION

USB

The KALEIDOSCOPE is powered by USB. Connect the USB Type-C and the device will start.

You can also use USB to:

- load and store presets [\[MANAGE PRESETS\]](#)
- load other firmware versions, including your own [\[FIRMWARE UPDATE\]](#)

■ MIDI

The KALEIDOSCOPE is a self-sufficient device. You can adjust the sound, as well as play tracks and run the sequencer using only the interface elements on the panel.

But KALEIDOSCOPE also has:

- TRS Type A MIDI IN,
- TRS Type A MIDI OUT.

Connect a MIDI keyboard or external sequencer to significantly expand the capabilities of the synthesizer. Details in the [\[GENERATIVE SEQUENCER\]](#)

By default, KALEIDOSCOPE tracks are set to MIDI channels 5, 6, 7 and 8, but you can freely change them [\[PARAMETER CFG-MIDI-VCH\]](#).

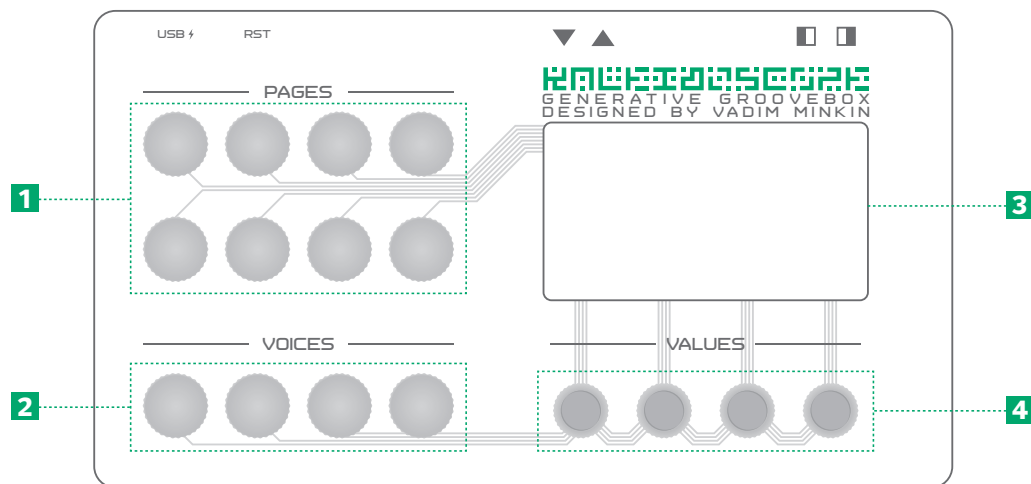
■ AUDIO

- Audio Out 1-2
- Audio Out 3-4

Two stereo 3.5mm jack connectors. Each of the 4 mono outputs can output the sum of any combination of voices, including individual tracks for subsequent processing by external effects chains. More details in the [\[PARAMETER MAST-SEN\]](#)

CONTROL

The KALEIDOSCOPE user interface is inspired by devices with page-based parameter organization. However, this interface has been rearranged, and received several new features that significantly improved the speed and quality of the user experience.



A picture of the KALEIDOSCOPE, which shows the front panel and labels all the elements

The user interface consists of 4 modules:

- 1 Page selector Page Selector Px (P1-P8)**
- 2 Track selector Track Selector Tx (T1-T4)**
- 3 Display**
- 4 Encoders for changing values Vx (V1-V4)**

■ DISPLAY

The display shows all the information about the synthesizer status. There are no other indicators on the panel that distract from work. The only place where the user can get information is the screen. Convenient, minimalistic, the eyes do not need to wander around the panel.

The peculiarity of the KALEIDOSCOPE display is that the screen always displays the current value of any parameter, taking into account all modulations. If a parameter changes for any reason, be it an envelope, aftertouch, modwheel or any combination of them, you will see its current value, which will change over time.

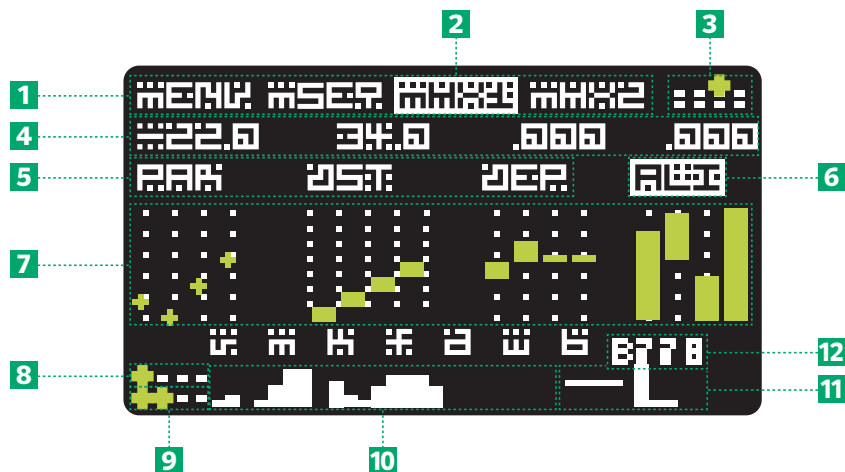
When working with a synthesizer, there is information that should always be in front of your eyes — this is the state of envelopes and modulators. The KALEIDOSCOPE display always shows thumbnails of all $4 * 7 = 28$ modulators, so that the user does not have moments of confusion:

- 4 volume envelopes,
- 4 modulation envelopes,
- 4 velocity envelopes,
- 4 aftertouch envelopes,
- 4 keytrack envelopes,
- 4 modwheels,
- 4 pitch bends.

KALEIDOSCOPE also tries to take into account the context. If your actions change the waveform, KALEIDOSCOPE will show the current waveform. When you are done editing, the waveform display is also turned off.

PAGE LAYOUT ON THE DISPLAY

The KALEIDOSCOPE OLED screen consists of 8 lines of 128 bytes. Each byte consists of 8 black and white pixels. A total of 8192 dots, which I tried to use wisely.



- 1** The names of the pages of the current parameter group
- 2** The currently selected page is highlighted by brightness inversion
- 3** The 8 sensors of the page group selector
- 4** The exact values of the currently edited parameter for each of the four tracks. When editing is not in progress, this line displays nothing.
 1. Most parameters have numeric values
 2. The modulation assignment has a parameter as its value. In this case, the value is equal to the parameter name.
 - On all *MX pages, when the encoders are not rotated, the PART parameter values are displayed for all tracks.
 - On all *ENV pages, when the encoders are not rotated, the PERI parameter values are displayed for all tracks.
 3. The patching parameters have no meaning, since the display already shows the module connection diagram.
- 5** The names of the parameters controlled by the encoder on the current page
- 6** The currently selected name is highlighted by brightness inversion
- 7** Approximate bar graphs with the values of 4 parameters for all 4 tracks. More details [\[VALUE SCALES\]](#)
- 8** Active tracks, whose parameters will be edited when the encoder is rotated
- 9** Held tracks, whose envelopes are held either by MIDI or from the track selector, or are in drone mode
- 10** Information about the value of all 28 modulators
- 11** Service Data
- 12** The last received MIDI message

▣▣ INCREMENTAL ENCODERS WITH A BUILT-IN BUTTON

4 encoders for the number of parameters located on the page. By rotating the encoder and pressing its button, the user changes the values of the synthesizer parameters.

▣▣ TRACK SELECTOR

4 sensors for the number of KALEIDOSCOPE tracks. These sensors are responsible for selecting the track with which the interaction will occur. You can select several tracks at once to change their values.

This selector also allows you to activate the track volume envelope, the track modulation envelope, and activate/deactivate the drone mode of envelopes.

▣▣ PAGE SELECTOR

8 sensors responsible for selecting a page with parameters in the KALEIDOSCOPE architecture.

▣▣ INTERACTING WITH THE INTERFACE

Any parameter can be changed using a simple algorithm:

1. Using the page selector, select the page on which the desired parameter is located. More information on selecting pages and locating parameters is provided in [\[PART 2\]](#) of this guide. You can also refer to the [\[SUMMARY TABLE OF PARAMETERS\]](#)
2. Using the track selector, select the track whose parameter you want to change. In KALEIDOSCOPE, it is possible to change the parameter of several voices at once. As soon as you touch the sensors, you immediately activate the volume envelopes of these tracks. Thus, you will be able to hear the result of your actions.



The behavior of the track selector sensors can be changed (for example, to disable envelope activation or to work in drone mode, when the voice can only be turned off by pressing it again). For more details, see [\[TSEL PARAMETER\]](#).



The track selector sensors can also be used simply to play the instrument. This way you can trigger envelopes. To change the frequency of the track oscillator, use the TUN parameter.

- After holding one or more track selector sensors, turn the encoder corresponding to the parameter. The parameter values of the selected tracks will change relatively. Turning the encoder will change the selected parameters by a certain value corresponding to the number of clicks made during the rotation.



If the encoder is pressed during the rotation, then for each click the values will change by 12 units.

This rule works for almost all parameters, except those related to patching.

For patching parameters, rotating the encoder while holding it will shift the patch.

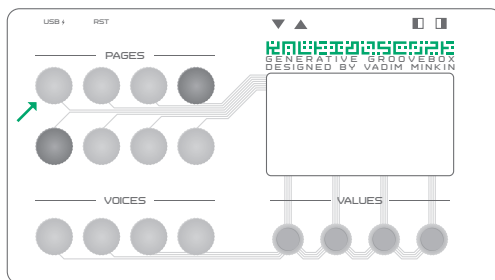


When changing a parameter, the current values will appear in the line above the parameter names.

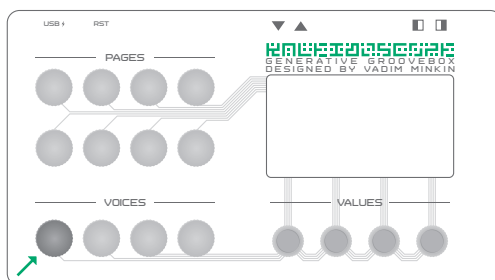


When changing timbre parameters related to the waveform of a signal, an image of the waveform of the sum of all currently playing tracks will appear on the screen. The waveform can also always be found on a separate SCOP page.

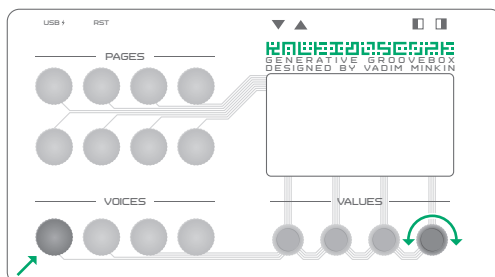
Select page



Select Track



Tweak encoder while holding



SUMMARY OF ALL OPTIONS FOR INTERACTION BETWEEN THE TRACK SELECTOR AND THE ENCODER

Touching the track sensor and then holding it in any mode (parameter CFG-INTF-TSEL) **activates** the track for subsequent editing of its parameters by the encoder.

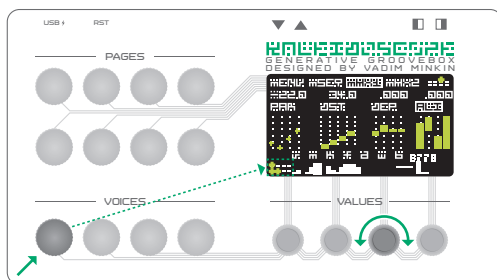
Releasing the track sensor in any mode (parameter CFG-INTF-TSEL) **deactivates** the track for editing by the encoder.

Rotating the encoder changes the value of the parameter assigned to it on the page of all **currently activated** tracks by 1 conventional unit per click.

Rotating the pressed encoder changes the value of the parameter assigned to it on the page of all **currently activated** tracks by 12 conventional units per click.



If no track is currently active, turning the encoder will show the current parameter values, but will not change them.



This is the basic principle that always works. Active voices are displayed in the upper row of the activation and hold indicator.

The method of starting the envelopes differs for the track selector sensor modes.

The track operates in HOLD mode (CFG-INTF-TSEL = 0):

- touching the track sensor and then holding it
- starts and holds the track volume envelope, as well as the modulation envelope
- releasing the sensor
- releases the track volume envelope and modulation envelope

The track operates in DRONE mode (CFG-INTF-TSEL = 1):

- quickly double-clicking the sensor
 - starts the track volume envelope and modulation envelope, if they were previously released
 - releases the track volume envelope and modulation envelope, if they were previously launched

Track works in EDIT mode (CFG-INTF-TSEL = 2):

- does not start or releases envelopes
- you can start an envelope in this mode only via MIDI

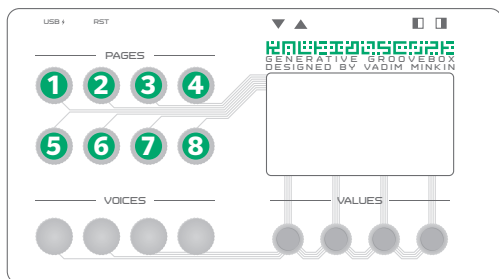
More details [\[TRACK SELECTOR MODES\]](#).

■ PAGE SELECTION

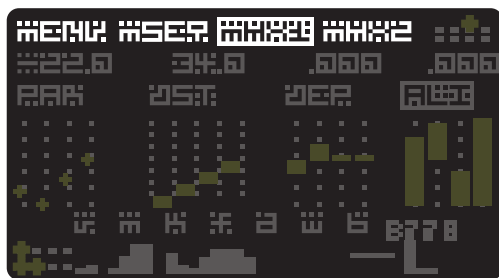
Each page contains only 4 parameters. But the values for all tracks are presented. That is, from one page you can change a total of $4 \times 4 = 16$ parameters.

The location of the parameters and their role in the synthesizer are described in detail in Part 2 of this guide.

In the **[PARAMETER SUMMARY TABLE]** you can always quickly find the desired parameter.



All Kaleidoscope pages are divided into 8 groups by the number of page selector sensors.



There are 4 pages inside each group.

The currently selected page is marked with brightness inversion.

The parameters are grouped so that you have to navigate the interface as little as possible.

To select a page with a parameter, you need to know its group and the page number within this group.

1. Press the sensor corresponding to the group.
2. On the opposite row of page selector sensors, press the sensor corresponding to the page within the group.

EXAMPLE 1

To select a page with the settings of the first destination of the mMX1 modulation envelope, you need to:

- Press and hold sensor 3. The names of all pages in this group will immediately appear in the top line of the display:
 - mENV,
 - mSEQ,
 - mMX1,
 - mMX2,
 - One of them will be highlighted with brightness inversion to show that you are currently on this page.
- Without releasing sensor 3, press sensor 7, corresponding to the 3rd page within the group — page mMX1.
- After that, page mMX1 will be highlighted with color inversion, and the corresponding parameters will be displayed on the page.

EXAMPLE 2

To select the page with the aENV channel aftertouch envelope settings, you need to:

- Press and hold sensor 5. The names of all pages in this group will immediately appear in the top line of the display:
 - aENV,
 - aMX,
 - fENV,
 - fMX,
 - One of them will be highlighted with brightness inversion to show that you are currently on this page.
- Without releasing sensor 5, press sensor 1, which corresponds to the 1st page inside the group — the aENV page.
- After that, the aENV page will be highlighted with color inversion, and the corresponding parameters will be displayed on the page.

If you do not know in which group of pages the desired page is, you can simply go through all 8 groups.

It will be found in one of them.



This method may seem more complicated than the simple cyclic page switching implemented in other synthesizers. But this method has a clear advantage. At any time, you can switch to any of the possible 32 pages of the synthesizer in 2 short actions.

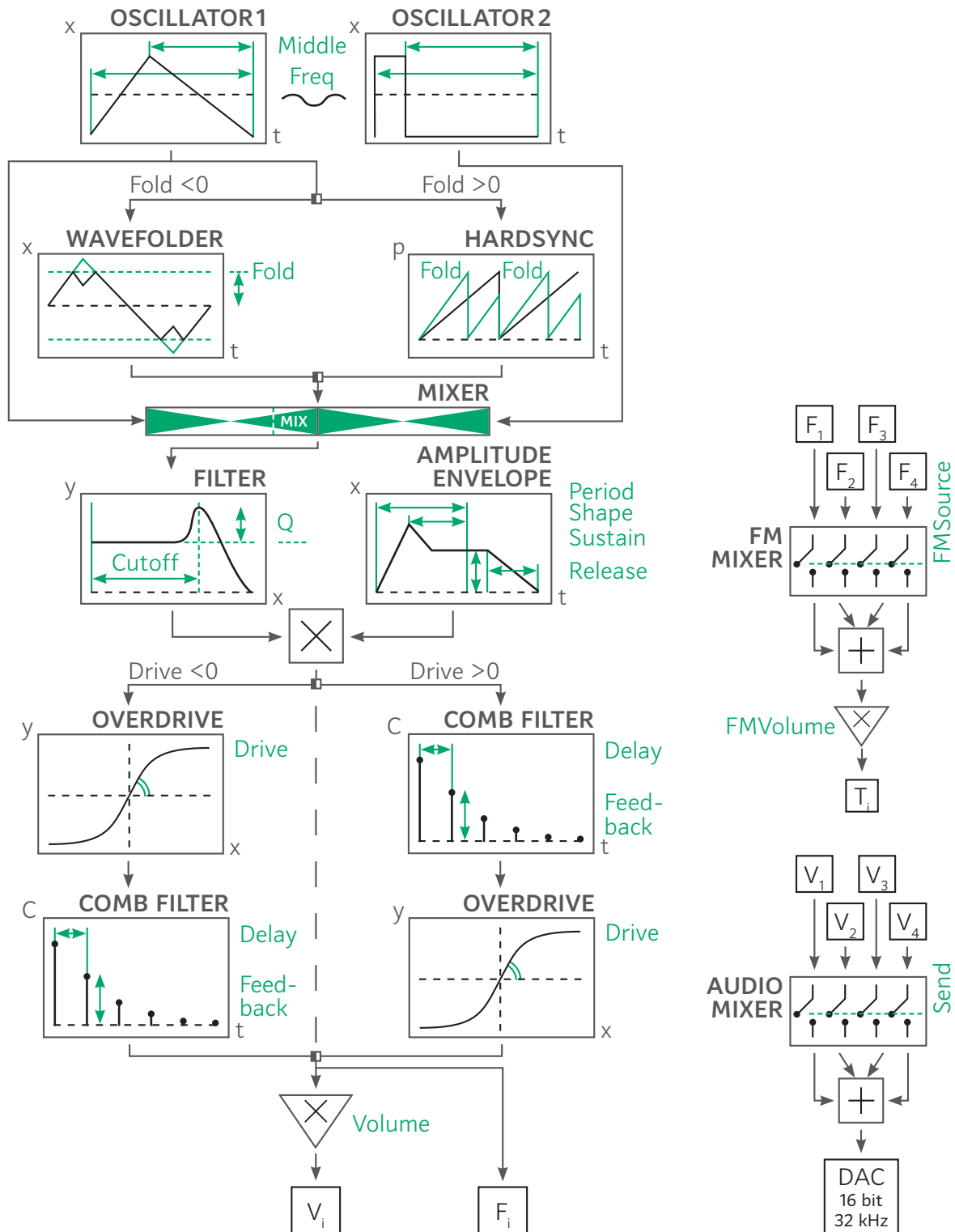
At the same time, the cyclic method requires additional mental resources to calculate the current position in the group, determining how many times you need to press the sensor to cyclically get to the desired location. This either slows down or leads to errors when you miss and have to go through another circle.

That's all you need to know about the KALEIDOSCOPE interface. No hidden menus, settings, or other "joys" of digital synthesizers. In KALEIDOSCOPE, everything is a parameter. From the MIDI channel to the memory cell number from which to load a preset.

To learn more about the location of parameters in the architecture, go to the next [\[PART 2\]](#).

PART 2. PARAMETERS AND INTERNAL STRUCTURE

The picture shows the audio path of the KALEIDOSCOPE.



All parameters of the Kaleidoscope are divided into groups.

Each group contains no more than 4 pages.



There are several empty groups (sensors 7 and 8). In future firmware versions, new parameters will appear in them.

Each page contains 4 parameters. Each parameter is controlled by its own encoder.

SUMMARY TABLE OF THE LOCATION OF PARAMETERS ON THE PAGES

Group	Page 1	Page 2	Page 3	Page 4
Waveform Control over the Spectrum 	ADTV Additive Parameters	SBTV Subtractive Parameters	FREQ Frequency Controls	MAST Volume, Audio routing and FM Synthesis
Volume Envelope Volume Envelope Controls 	vENV Shape Controls	vSEQ Generative Sequencer Controls	vMX1 Modulation Destination 1 Controls	vMX2 Modulation Destination 2 Controls
Modulation Envelope Modulation Envelope Controls 	mENV Shape Controls	mSEQ Generative Sequencer Controls	mMX1 Modulation Destination 1 Controls	mMX2 Modulation Destination 2 Controls
Configuration Miscellaneous 	MIDI MIDI Channels Controls	INTF Track Selector and BPM	PRST Preset Management	SCOP Waveform and Spectrum Scopes
External Modulators 1 MIDI Modulators Controls 1 	aMX Aftertouch Modulation Destination Controls	aSHP Aftertouch Smoothing Controls	fMX Velocity Modulation Destination Controls	fSHP Velocity Smoothing Controls
External Modulators 2 MIDI Modulators Controls 2 	kMX Keytrack Modulation Destination Controls	kSHP Keytrack Smoothing Controls	wMX Mod Wheel Modulation Destination Controls	pMX Pitch Bend Modulation Destination Controls

Below is a detailed description of all the parameters, pages and page groups of KALEIDOSCOPE.

WAVEFORM GROUP

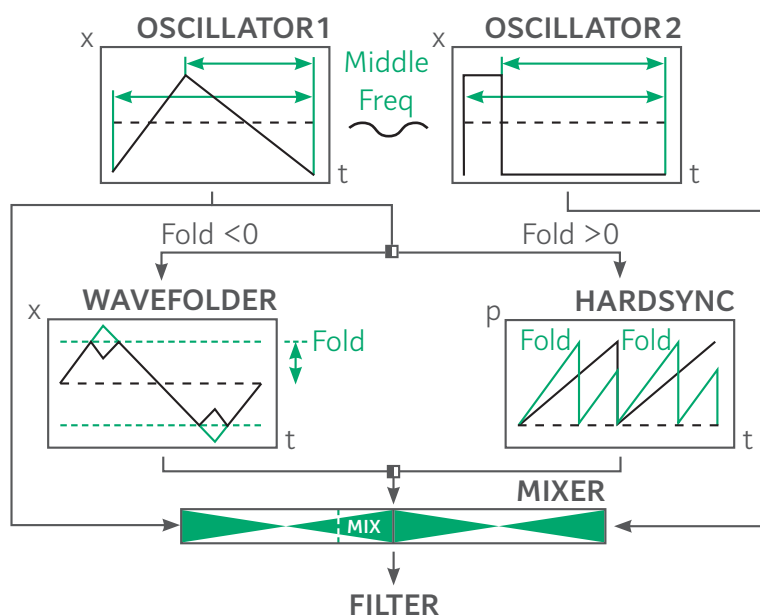
This group deals with the spectral composition of the sound and the waveform control.



ADTV PAGE

This page deals with the waveform. Additive parameters that affect the composite oscillator modules and the over-drive module.

ARCHITECTURE



The KALEIDOSCOPE oscillator generates two connected waves simultaneously:

- asymmetrical triangle,
- pulse.

Both of these waveforms have different harmonic composition and brightness of timbre.

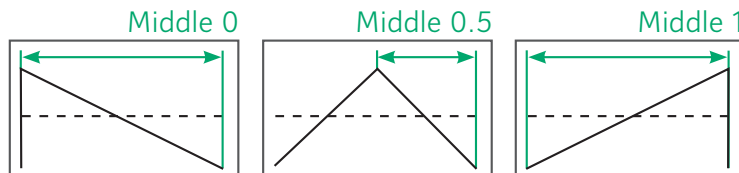
The signal of the asymmetrical triangle is fed to the input of the wavefolder. The wavefolder output can then be mixed with either the original asymmetrical triangle or the pulse.

The output is a great variety of different waveforms and timbres. And all this is controlled by only 3 parameters.

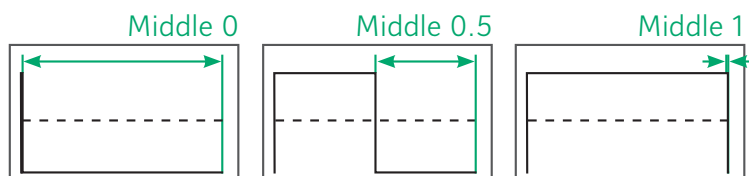
WAVEFORMS MIDL PARAMETER (MIDDLE)

VALUE FROM: 0F:1F SCALE: BIDIRECTIONAL

MIDL controls the position of the midpoint within one period of the oscillator.



By modulating MIDL, the triangle turns into a saw, something similar to the waveform of a guitar string is obtained.



By changing the pulse length, a PWM signal is obtained, which has a very rich sound.

WAVEFORMS FOLD PARAMETER (FOLD)

VALUE FROM: -1F:1F SCALE: BIDIRECTIONAL

The wavefolder has two different modes:

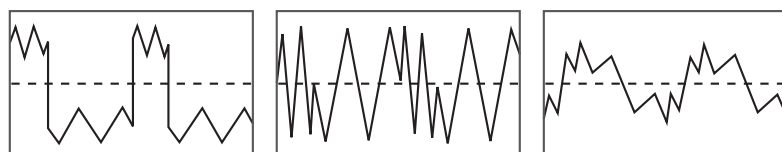
- fold > 0 Wavefolder works as hardsync by multiplying oscillator phase,
- fold < 0 Wavefolder folds waveform corresponding to fold offset level.

! The extreme values of the FLD parameter, the number of folds reaches about a million. Any signal turns into noise-like.

WAVEFORMS MIX PARAMETER (MIXER)

VALUE FROM: -1F:1F SCALE: BIDIRECTIONAL

In some sense, the wavefolder works as a frequency multiplier, so it suppresses low-frequency harmonics of the oscillator.



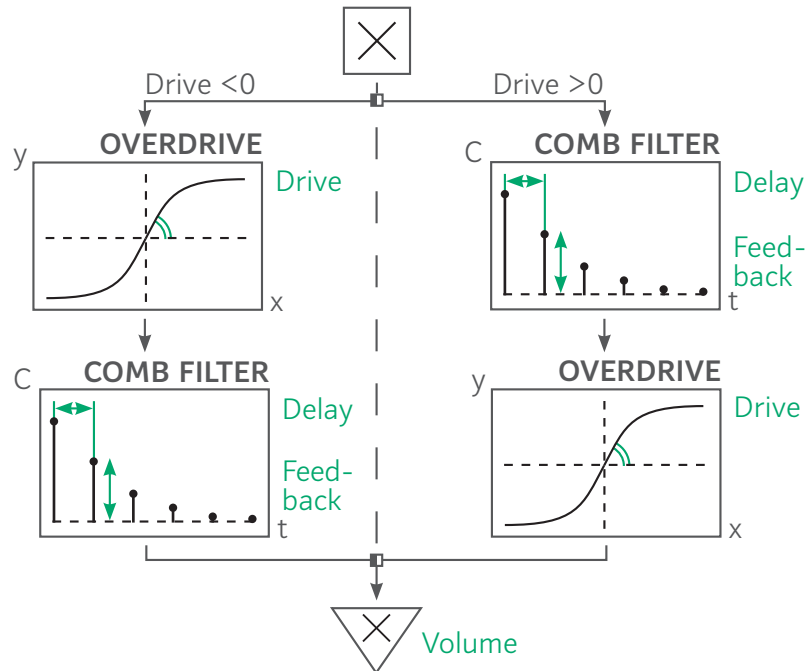
The MIX parameter allows you to bring back low-frequency harmonics by mixing the wavefolder output with one of the oscillator signals:

- asymmetrical triangle,
- pulse.

DRIV PARAMETER (DRIVE)

VALUE FROM: 0f:1f SCALE: BIDIRECTIONAL

ARCHITECTURE



Enlarges the place where the overdrive unit may be present.

The overdrive module is located after the resonant filter.

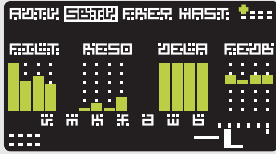
Can act softly for a little saturation. Can act hard in the fuzz style, turning any incoming signal into a square one.

In classic overdrive modules, the amount of distortion increases due to the amplification of the amplitude of the incoming signal. The higher the amplitude, the greater the part of the signal experiences distortion.

In KALEIDOSCOPE, the overdrive works differently. It smoothly changes its shape from soft to uncompromisingly hard.

DRIV has two different modes:

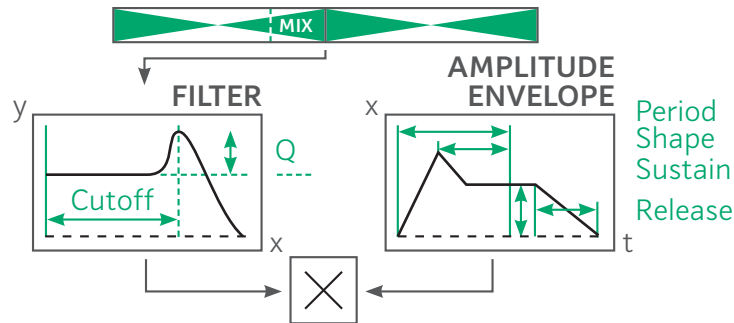
- after the volume envelope. The shape of the distorted signal (and therefore the timbre of the sound) depends on the current value of the volume envelope. The higher the volume, the stronger the distortion. In other words, the timbre changes along with the volume change.
- after the comb filter output. The sound timbre depends on the current value of the volume envelope, but also on the combo filter parameters. The overdrive becomes even more dense and dirty.



SBTV PAGE

On this page, the parameters control the filtering of the signal. Subtractive parameters that affect the filter with resonance and the comb filter.

ARCHITECTURE



Enlarging the filter module from the original architecture

The output of the oscillator mixer is fed to the input of a resonant filter, which is capable of cutting out some of the harmonics. The filter can operate in low-pass or high-pass mode. The filter is also capable of amplifying a small area near the cutoff frequency due to resonance.

FILT: FILT PARAMETER (FILTER)

VALUE FROM: C(-1):G9 SCALE: RISING

Cutoff frequency of the resonant low-pass filter.

Tuned to the frequencies of midi notes.

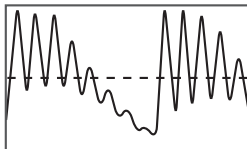
This is useful when you want the resonance to emphasize a certain harmonic, spaced from the base harmonic by a certain interval.

RESO: RESO PARAMETER (RESONANCE)

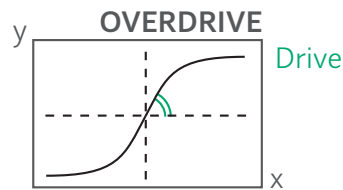
VALUE FROM: 0f:1f SCALE: RISING

Q — quality factor of the oscillatory circuit. Filter resonance.

At high resonance values the filter is capable of self-oscillation.



ARCHITECTURE



The filter output goes through amplification, and then overdrive (the position of the overdrive in the audio path depends on its settings). After that, the signal goes to the input of the comb filter, which is capable of cutting out some of the harmonics in accordance with a periodic pattern. Its frequency response is comb-shaped, which allows you to create complex modifications of the spectrum.

! If you set the comb filter delay length equal to the period of the incoming signal, then no change in the spectrum will occur — the signal harmonics will fall exactly on the peaks of the filter's frequency response. But if the comb filter delay length and the signal period are out of sync, then the filter will cut out some of the harmonics, transforming the sound.

The comb filter is used:

- to create resonant overtones
- to add echo
- to synthesize string sounds using the Karplus-Strong method. This is just one of the methods of synthesis of physical modeling.

! Due to the presence of an internal saturation module, self-excitations are possible, which depend on the delay length and the feedback coefficient of the comb filter.

DELA PARAMETER (DELAY)

VALUE FROM: C(-1):G9 SCALE: FALLING

The delay length of the comb filter.

The values 0:35 are set in a rhythmic sense. The period between echoes is tied to the track tempo (BPM parameter). The period is shortened by a factor of 2 for every 12 clicks of the encoder. The values 36:127 are set in a harmonic sense. The delay length in this case is set to the frequencies of the corresponding MIDI notes, so that the combo filter can be played note by note.

The shorter the delay time, the further the peaks in the frequency domain are from each other. With a delay length of 1 sample, such a filter turns into a regular Low Pass.

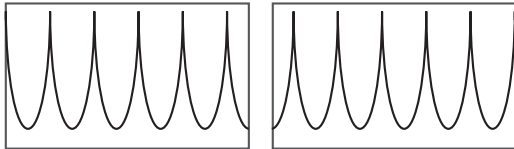
FEDB PARAMETER (FEEDBACK)

VALUE FROM: -1F:1F SCALE: BIDIRECTIONAL

Echo decay time in the delay length.

Bidirectional parameter.

With positive values of the parameter, the frequency response of the filter has dips in even half-periods of the spectrum.



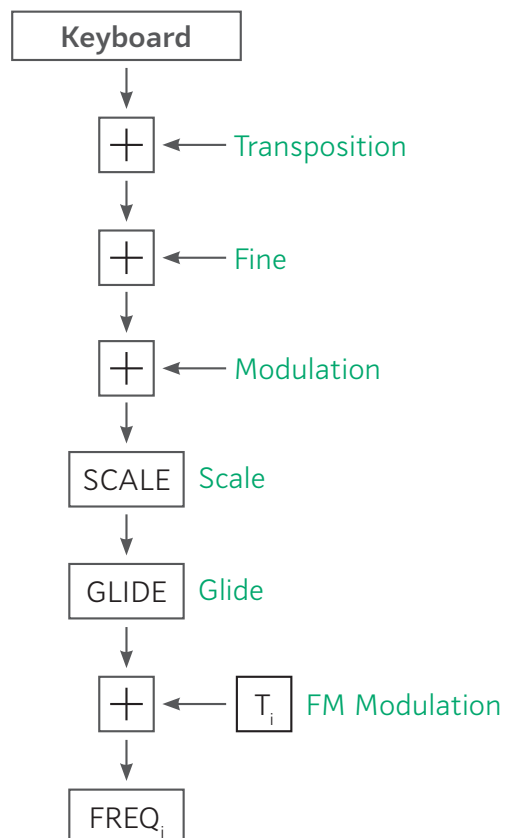
With negative values of the parameter, the frequency response of the filter has dips in odd half-periods of the spectrum.



FREQ PAGE

On this page, the parameters concern the control of the oscillator frequency.

ARCHITECTURE



Incoming MIDI notes are first transposed by the selected number of semitones — TRAN (Transposition). After this, the frequency is fine-tuned within a semitone — FINE (Fine Tuning). Then the modulation value from the envelopes and MIDI modulators is added. Then the frequency is quantized — SCALE (SCALE). Then the previous frequency begins to smoothly transition to the new one under the action of the glide — GLIDE (Glide). Then the FM distortion is added to the frequency — Ti .

TRAN PARAMETER (TRANPOSE)

VALUE FROM: -63i:64i SCALE: BIDIRECTIONAL

Transposition of the voice oscillator frequency.



When the device starts, each voice is initiated with the frequency of the note C2. Then the note is transposed and the oscillator frequency is set to the corresponding value.

FINE PARAMETER (FINE)

VALUE FROM: -1f:1f SCALE: BIDIRECTIONAL

Fine frequency adjustment within a semitone up or down.

GLID PARAMETER (GLIDE)

VALUE FROM: 10f:0.001f SCALE: FALLING

Speed of oscillator frequency tuning.

SCAL PARAMETER (QUANTIZE)

VALUE FROM: 0f:1f SCALE: ONE-HOT

Quantization of the oscillator frequency value for some scales.

Available values:

OFF — quantization disabled

nMAJ — natural major

nMIN — natural minor

PENT — major pentatonic

HEXA — hexatonic

BLUE — blues hexatonic

EMP — empty scale. Will be filled in future firmware

FIX — the oscillator ignores the value of MIDI notes. The frequency is adjusted only by the TRAN parameter (transposition).



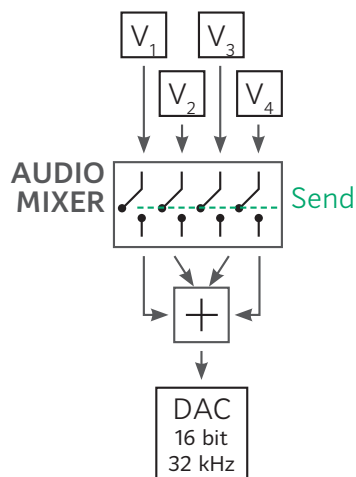
If the scale is not OFF or FIX, then TRAN determines the key of the scale.



MAST PAGE

This page contains parameters related to the control of the Audio Mixer-Router, FM Mixer-Router, and FM synthesis parameters.

ARCHITECTURE



The output of the comb filter (or overdrive, if it is configured to work behind the combo filter) is fed to the input of the Audio Mixer-Router.

KALEIDOSCOPE has 4 mono audio outputs. The sum of any combination of track outputs can be sent to each of them.

VOLU PARAMETER (VOLUME)

VALUE FROM: -1F:1F SCALE: BIDIRECTIONAL

Voice volume.

Positive values of the parameter determine the coefficient by which the audio output of the track is multiplied before being fed to the audio DAC.

Negative volume values activate the Offset mode.

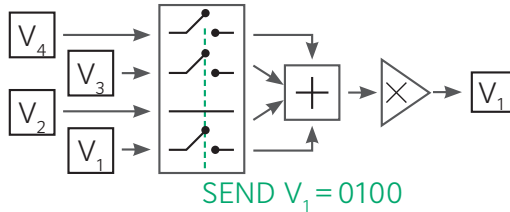
In this mode, the signal volume is constant and maximum.

The VOLU parameter controls the signal offset relative to 0 in the range from -2 to 2. So it also controls the volume, but in an experimental manner.

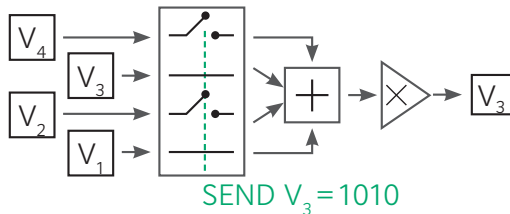
SEND PARAMETER (SEND)

VALUE FROM: 0i:15i SCALE: 4 BITS

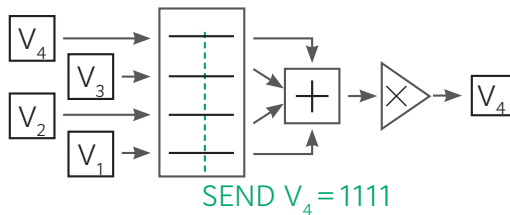
Routing the track output to 4 mono audio outputs.



This means that the signal at audio output 1 consists only of the audio output of track 2.



This means that the signal at audio output 3 consists of the sum of audio track 1 and track 3.



This means that the signal at audio output 4 consists of the sum of the audio of all four tracks.

The output of the comb filter (or overdrive, if it is configured to work behind the combo filter) goes to the input of the FM Mixer-Router.

Each track of the Kaleidoscope has an input for frequency modulation. The sum of any combination of track outputs can be sent to the input of each track.

FMVO PARAMETER (FM VOLUME)

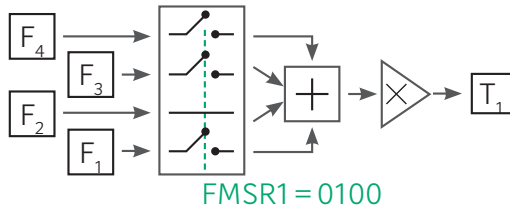
VALUE FROM: -1F:1F SCALE: BIDIRECTIONAL

Control over the amplitude of the incoming signal modulating the track frequency. Negative values correspond to signal subtraction.

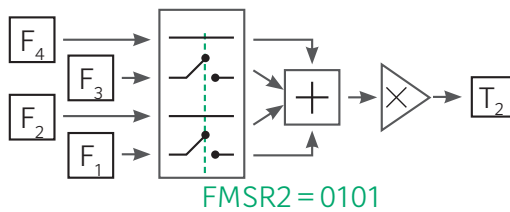
FMSR PARAMETER (FM SOURCE)

VALUE FROM: 0i:15i SCALE: 4 BITS

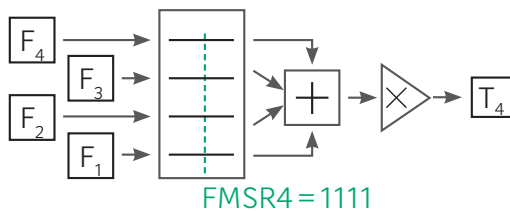
Selecting sources of FM modulation of the track.



This means that the frequency modulation signal of track 1 consists of the audio of track 2 only.



This means that the frequency modulation signal of track 2 is the sum of the audio of track 2 and track 3.



This means that the frequency modulation signal of track 4 is the sum of the audio of all tracks.

VOLUME ENVELOPE GROUP

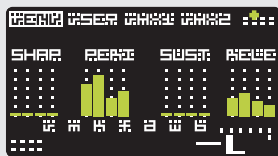
In this group, all parameters concern the volume envelope.

Each track has its own volume envelope.

You can start/release the envelope by pressing/releasing the corresponding track selector sensor. Detailed information in the **[TSEL PARAMETER)]**.

You can start envelopes from the keyboard simultaneously by setting them to a common MIDI channel.

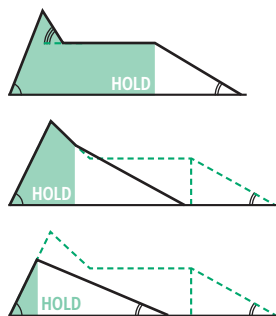
More details in the **[GENERATIVE SEQUENCER]**.



vENV PAGE

Controlling the shape of the volume envelope.

ARCHITECTURE



The KALEIDOSCOPE volume envelope has 4 stages:

- attack,
- decay,
- sustain,
- release.

In KALEIDOSCOPE, the emphasis is on the convenience of using the envelope in cyclic mode. Therefore, instead of separately controlling the attack and decay speeds, these stages are considered as parts of one asymmetrical triangular waveform, similar to an oscillator. As a result, the parameters that control the shape are:

- SHAP (Shape), similar to the MIDL (Middle Point) parameter of the oscillator
- PERI (Period), similar to the FREQ (Frequency) parameter of the oscillator

This choice will become clear to you when you get acquainted with the possibilities of looping the envelope and thus creating generative rhythmic sequences. Such a sequencer would not be so musical if it retained independent attack and decay parameters. More details in the **[GENERATIVE SEQUENCER]**.

SHAP SHAP PARAMETER (SHAPE)

VALUE FROM: 0F:1F SCALE: BIDIRECTIONAL

Shape of the asymmetrical attack-decay triangle.

PERI PERI PARAMETER (PERIOD)

VALUE FROM: 64:0.041 SCALE: RISING

Period of the attack-decay triangle.

! The period values are tied to the current BPM value. Once you have created dynamics, you will not lose them when changing the tempo.

Changing the value by 12 units changes the envelope period length by 2 times (per rhythmic octave). Within the rhythmic octave, the values are also tied to different beats of the bar.

- eighth,
- triplet,
- dotted note,
- pentole,
- septole,
- and others.

PERI is tied to the set BPM. When changing the BPM, the dynamics are not disturbed. The synthesizer adjusts the parameters automatically.

! The values of the period (PERI), release (RELE), rise (RISE) and fall (FALL) parameters correspond to the inverse values in fractions of a bar at the current BPM value.
 For example, a period value of 64 corresponds to $1/64$ of a bar.
 For example, a period value of 0.25 corresponds to $1/0.25 = 4$ bars.
 For example, a period value of 7 corresponds to $1/7$ of a bar.
 And so on.

SUST SUST PARAMETER (SUSTAIN)

VALUE FROM: 0F:1F SCALE: RISING

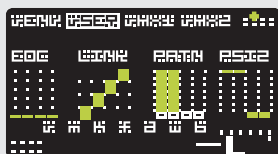
Sustain level control.

This parameter also displays the current volume envelope level on the display.

RELE RELE PARAMETER (RELEASE)

VALUE FROM: 1/64F:8F SCALE: RISING

The decay time of the volume envelope level after it is deactivated.

**vSEQ PAGE**

Page with settings for the generative sequencer of the volume envelope.

Each envelope has its own generative sequencer.

In general, this module was not created as a sequencer, but as an unusual way to create more complex envelope shapes.



Initially, an option was added to turn on/off the cyclicity of the envelope, as it was done in West Coast synthesizers.

But then I added ideas from my Ornament-8 synthesizer, when the envelope can launch not only itself, but also other envelopes, if they are not currently used by anyone.

The principle of its operation is given below.

Each envelope at the end of its attack-decay cycle generates an end of cycle (EOC) trigger. This trigger can launch any envelope of the same type.

The LINK (Link) parameter is responsible for selecting the envelopes to which the EOC trigger will be sent.

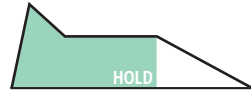
The EOC (End of Cycle) parameter enables or disables this feature.



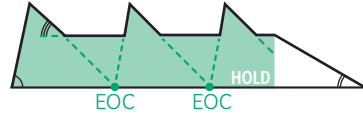
When the EOC trigger of one envelope launches another, the note of the launching voice is also transmitted to it. Taking into account the transposition, this allows you to use the second voice as a sub-oscillator.

ARCHITECTURE

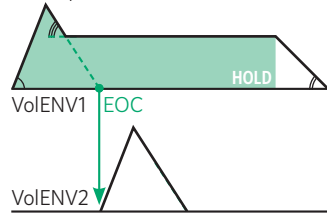
EOC=0



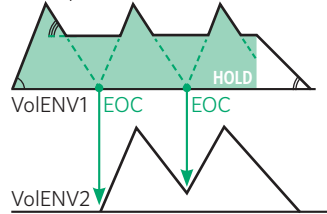
EOC=1
LNK₁=1



EOC=1
LNK₁=2



EOC=1
LNK₁=1&2



Additionally, KALEIDOSCOPE has the ability to mute the envelope on certain cycles. The PATN (Pattern) and PSIZ (Pattern Size) parameters..

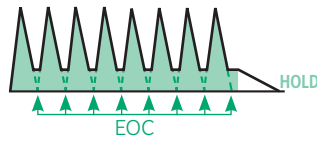
By creating different settings for different envelopes, you can create complex modulations and polyrhythmic patterns on the fly.

More details in the [\[GENERATIVE SEQUENCER\]](#) section.

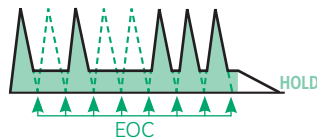
EOC=0



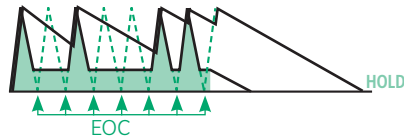
EOC=1 PTN=11111111
LNK₁=1



EOC=1 PTN=10100111
LNK₁=1



EOC=1 PTN=10100111
LNK₁=1



EOC PARAMETER (END OF CYCLE)

VALUE FROM: 0i:3i SCALE: 4 BITS

Selects the trigger source for restarting envelopes.

- off
- end of cycle (EOC).
 - This trigger is a virtual trigger that is sent to all envelopes selected by the LINK (Link) parameter at the end of the envelope decay stage.
 - If an envelope triggers itself, it simply goes into the attack stage and the process repeats cyclically as long as the envelope is held in the active state.



If an attempt is made to activate an envelope that is currently in a deactivated state (not used by other tracks), then this envelope goes into the attack phase.

The oscillator on the track of this envelope is sent the note of the activating track.



If the envelope works in cyclic mode, i.e. the EOC parameter = 1, the RELE parameter plays the role of the decay time. Thus, the PERI parameter controls the tempo of the envelope movement along the pattern mask, and the release controls the dynamics of the sound.

LINK PARAMETER (LINK)

VALUE FROM: 0i:15i SCALE: 4 BITS

Selects the envelopes to which the trigger defined in the TRIG parameter will be transmitted.

PATN PARAMETER (PATTERN)

VALUE FROM: 1i:255i SCALE: 8 BITS

Pattern that determines the shape of the envelope while it holded.



When the encoder is rotated with the track selector sensor pressed, the pattern mask value changes by 1 unit.

If the encoder is pressed, the pattern mask shifts. Bits that go beyond the boundaries are erased.

More details in the [\[GENERATIVE SEQUENCER\]](#).

PSIZ PARAMETER (PATTERN SIZE)

VALUE FROM: 1i:8i SCALE: ONE-HOT

Pattern length. From 1 to 8 steps.



If EOC = off, the pattern is ignored. The envelope works in the normal ADSR mode.

If EOC = on, the EOC trigger is generated according to the PERI parameter. But the envelope decays according to the RELE parameter.



vMX1 & vMX2 PAGES

Volume envelope modulation matrix.
In addition to volume control, this envelope can modulate 2 parameters. Each parameter has its own settings page.

PARAM PARM PARAMETER (PARAMETER)

VALUE FROM: 1i:32i SCALE: ONE-HOT

Selects the parameter to modulate.

DEST DEST PARAMETER (DESTINATION)

VALUE FROM: 1i:4i SCALE: ONE-HOT

Destination track number.

DEPT DEPT PARAMETER (DEPTH)

VALUE FROM: -10f:1f SCALE: BIDIRECTIONAL

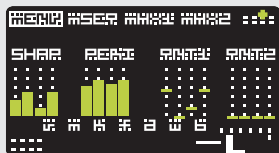
Bidirectional depth of action.

ALIA ALIA PARAMETER (ALIAS) VALUE FROM/SCALE: DEPENDS ON THE SELECTED PARAMETER

This parameter allows you to tweak the modulated parameter without leaving the modulation page. That is, the one that is changed by the modulator.
This is very convenient, because you do not need to leave the modulator page to change the base value of the parameter.

MODULATION ENVELOPE GROUP

In this group, all parameters concern the control of the general-purpose modulator.



mENV PAGE

The shape of the modulator envelope has only two stages — attack and decay.

These stages are controlled in the same way as the volume envelope via the Shape and Period parameters. More details in the [\[GENERATIVE SEQUENCER\]](#) section.

SHAP SHAP PARAMETER (SHAPE)

VALUE FROM: 0F:1F SCALE: BIDIRECTIONAL

Shape of the asymmetrical attack-decay triangle.

PERI PERI PARAMETER (PERIOD)

VALUE FROM: 64:0.041 SCALE: RISING

Period of the attack-decay triangle. The period values are tied to the current BPM value.

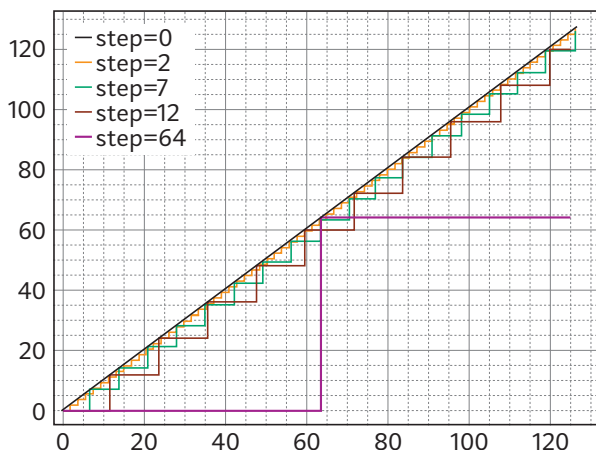
Changing the value by 12 units changes the duration of the envelope period by 2 times (by an octave, but in the rhythmic sense). Within the rhythmic octave, the values are also tied to different beats of the bar:

- eighth,
- triplet,
- dotted note,
- pentole,
- seventole and others.

QNT1 QNT1 PARAMETER (QUANTISE OF DESTINATION1)

VALUE FROM: 0:127 SCALE: RISING

Quantization of the modulator output signal for destination 1. The linear signal of the modulator turns into a stepped signal. A particularly interesting value is 12. Allows you to change the selected parameter by octaves.



Available values:

- 0 — no quantization,
- 1 — quantization by semitones,
- 2 — quantization by tones,
-
- 12 — quantization by octaves,
-
- 24 — quantization by 2 octaves,
- 30 — mSEQ quantization by modulator pattern mask values. If mask 1—modulator value 1. If mask 0 — modulator value 0,
- 31 — vENV modulator value duplicates track volume envelope value. This adds additional destination to volume envelope.

QNT2 PARAMETER (QUANTISE OF DESTINATION2) **VALUE FROM: 0:127** **SCALE: RISING**

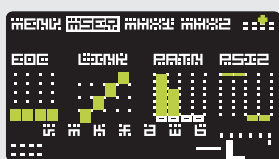
Quantization of the modulator output signal for destination 2.



This parameter allows you to implement an arpeggiator.

The arpeggiator can change not only the oscillator frequencies, but also the frequency of the filter with resonance and the delay of the comb filter, whose values are tied to the values of midi notes.

If you use quantize the modulation of the envelope period, you can create various rhythmic effects associated with doubling the rhythm.



mSEQ PAGE

Page with settings for the generative sequencer of the modulation envelope.

Each envelope has its own generative sequencer. The modulation envelope sequencer works the same way as the volume envelope sequencer.

More details in the **[GENERATIVE SEQUENCER]** section.

EOC PARAMETER (END OF CYCLE)

VALUE FROM: 0i:3i **SCALE: 4 BITS**

Selects the trigger source for restarting envelopes.

- ☐ off
- ☒ end of cycle (EOC).
 - This trigger is a virtual trigger that is sent to all envelopes selected by the LINK (Link) parameter when the envelope's decay stage ends.
 - If an envelope triggers itself, it simply goes into the attack stage and the process repeats itself while the envelope is held in the active state.



If an attempt is made to activate an envelope that is currently in a deactivated state (not used by other tracks), this envelope goes into the attack phase.

LINK PARAMETER (LINK)

VALUE FROM: 0i:15i **SCALE: 4 BITS**

Selects the modulation envelopes to which the trigger defined in the EOC parameter will be transmitted.

PATN PATN PARAMETER (PATTERN)

VALUE FROM: 1i:255i SCALE: 8 BITS

Pattern that determines the restart of the envelope.



When the encoder is turned with the sensor pressed, the pattern mask value changes by 1 unit. If the encoder is pressed, the pattern mask is shifted. Bits that go beyond the limits are erased.

PSIZ PSIZ PARAMETER (PATTERN SIZE)

VALUE FROM: 1i:8i SCALE: ONE-HOT

Pattern length. From 1 to 8 steps.



If EOC = off, the pattern is ignored.
The envelope works in the usual AD mode.



mMX1 и mMX2 PAGES

Modulation matrix of the modulation envelope. The modulation envelope has two destinations with an assignable parameter and a track to which this parameter belongs.

PARM PARM PARAMETER (PARAMETER)

VALUE FROM: 1i:32i SCALE: ONE-HOT

Selects the parameter for modulation.

DEST DEST PARAMETER (DESTINATION)

VALUE FROM: 1i:4i SCALE: ONE-HOT

Destination track number.

DEPT DEPT PARAMETER (DEPTH)

VALUE FROM: -1f:1f SCALE: BIDIRECTIONAL

Bidirectional depth of effect. Intensity.

ALIA ALIA PARAMETER (ALIAS) VALUE FROM/SCALE: DEPENDS ON THE SELECTED PARAMETER

This parameter allows you to tweak the modulated parameter without leaving the modulation page.

GROUP XMOD1

Group of external modulation sources. It is used to set up modulation by channel aftertouch messages and velocity envelope.



aMX PAGE

Channel aftertouch modulation matrix.

PARAM PARM PARAMETER (PARAMETER)

VALUE FROM: 1i:32i SCALE: ONE-HOT

Selecting a parameter for modulation.

DEST DEST PARAMETER (DESTINATION)

VALUE FROM: 1i:4i SCALE: ONE-HOT

Track number of the destination.

DEPT DEPT PARAMETER (DEPTH)

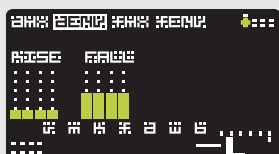
VALUE FROM: -1f:1f SCALE: BIDIRECTIONAL

Bidirectional depth of action. Intensity.

ALIA ALIA PARAMETER (ALIAS) VALUE FROM/SCALE: DEPENDS ON THE SELECTED PARAMETER

This parameter allows you to tweak the modulated parameter without leaving the modulation page. That is, the one that is changed by the modulator.

This is very convenient because you don't have to leave the Modulator page to change the base value of the parameter.



aENV PAGE

Channel aftertouch smoothing parameters.

RISE RISE PARAMETER (RISE)

VALUE FROM: 0f:1f SCALE: RISING

Smoothing speed when the parameter changes rising.

FALL FALL PARAMETER (FALL)

VALUE FROM: 0f:1f SCALE: RISING

Smoothing speed when the parameter changes falling.



FMX PAGE

Матрица модуляции скорости нажатия (f is for force).

PARAM PARM PARAMETER (PARAMETER)

VALUE FROM: 1i:32i SCALE: ONE-HOT

Selecting a parameter for modulation.

DEST DEST PARAMETER (DESTINATION)

VALUE FROM: 1i:4i SCALE: ONE-HOT

Track number of the destination.

DEPT DEPT PARAMETER (DEPTH)

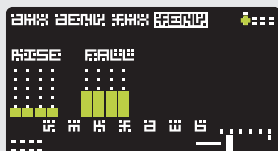
VALUE FROM: -1f:1f SCALE: BIDIRECTIONAL

Bidirectional depth of action. Intensity.

ALIA ALIA PARAMETER (ALIAS) VALUE FROM/SCALE: DEPENDS ON THE SELECTED PARAMETER

This parameter allows you to tweak the modulated parameter without leaving the modulation page. That is, the one that is changed by the modulator.

This is very convenient because you don't have to leave the Modulator page to change the base value of the parameter.



FENV PAGE

Velocity smoothing parameters. Allows you to turn velocity into an additional AHR envelope.

RISE RISE PARAMETER (RISE)

VALUE FROM: 0f:1f SCALE: RISING

Smoothing speed when the parameter changes rising.

FALL FALL PARAMETER (FALL)

VALUE FROM: 0f:1f SCALE: RISING

Smoothing speed when the parameter changes falling.

XMOD2 GROUP

External modulation source group. Keytrack, modwheel and pitchbend modulation are configured here.



kMX PAGE

Keytrack modulation matrix.

PARM PARM PARAMETER (PARAMETER)

VALUE FROM: 1i:16i SCALE: ONE-HOT

Selecting a parameter for modulation.

DEST DEST PARAMETER (DESTINATION)

VALUE FROM: 1i:4i SCALE: ONE-HOT

Track number of the destination.

DEPT DEPT PARAMETER (DEPTH)

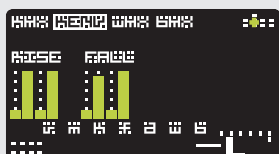
VALUE FROM: -1f:1f SCALE: BIDIRECTIONAL

Bidirectional depth of action. Intensity.

ALIA ALIA PARAMETER (ALIAS) VALUE FROM/SCALE: DEPENDS ON THE SELECTED PARAMETER

This parameter allows you to tweak the modulated parameter without leaving the modulation page. That is, the one that is changed by the modulator.

This is very convenient because you don't have to leave the Modulator page to change the base value of the parameter.



kENV PAGE

Keytrack smoothing parameters. Allows you to implement a parametric glide for the modulated parameter.

RISE RISE PARAMETER (RISE)

VALUE FROM: 0f:1f SCALE: RISING

Smoothing speed when the parameter changes rising.

FALL FALL PARAMETER (FALL)

VALUE FROM: 0f:1f SCALE: RISING

Smoothing speed when the parameter changes falling.



pMX PAGE

Pitch bend modulation matrix.

PARAM PARM PARAMETER (PARAMETER)

VALUE FROM: 1i:16i SCALE: ONE-HOT

Selecting a parameter for modulation.

DEST DEST PARAMETER (DESTINATION)

VALUE FROM: 1i:4i SCALE: ONE-HOT

Track number of the destination.

DEPT DEPT PARAMETER (DEPTH)

VALUE FROM: -1f:1f SCALE: BIDIRECTIONAL

Bidirectional depth of action. Intensity.

ALIA ALIA PARAMETER (ALIAS) VALUE FROM/SCALE: DEPENDS ON THE SELECTED PARAMETER

This parameter allows you to tweak the modulated parameter without leaving the modulation page. That is, the one that is changed by the modulator.

This is very convenient because you don't have to leave the Modulator page to change the base value of the parameter.



wMX PAGE

Modulation matrix modulation wheel.

PARAM PARM PARAMETER (PARAMETER)

VALUE FROM: 1i:16i SCALE: ONE-HOT

Selecting a parameter for modulation.

DEST DEST PARAMETER (DESTINATION)

VALUE FROM: 1i:4i SCALE: ONE-HOT

Track number of the destination.

DEPT DEPT PARAMETER (DEPTH)

VALUE FROM: -1f:1f SCALE: BIDIRECTIONAL

Bidirectional depth of action. Intensity.

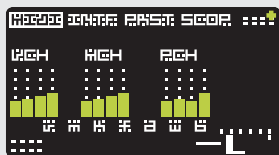
ALIA ALIA PARAMETER (ALIAS) VALUE FROM/SCALE: DEPENDS ON THE SELECTED PARAMETER

This parameter allows you to tweak the modulated parameter without leaving the modulation page. That is, the one that is changed by the modulator.

This is very convenient because you don't have to leave the Modulator page to change the base value of the parameter.

CFG GROUP

This group contains all parameters related to the configuration of the KALEIDOSCOPE interface.

**MIDI PAGE**

Configuring MIDI channels for envelopes.
 Details in [\[POLYPHONY AND MIDI CHANNELS\]](#).
 Details in [\[TRACK SELECTOR MODES\]](#).

VCH PARAMETER**VALUE FROM: 1i:32i SCALE: ONE-HOT**

Input MIDI channel, noteOn/noteOff messages from which start/release the volume envelope.

MCH PARAMETER**VALUE FROM: 1i:32i SCALE: ONE-HOT**

Input MIDI channel, noteOn/noteOff messages from which start/release the modulation envelope.

PCH PARAMETER**VALUE FROM: 1i:16i SCALE: ONE-HOT**

Input MIDI channel for synth configuration from external MIDI controller, which accepts parameters (CC messages).

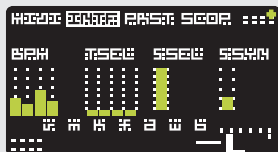


In classic synthesizers, all three of these channels are the same for each voice. But in the KALEIDOSCOPE there is complete freedom, you may want to combine several voices into one voice by combining envelopes (set them to one MIDI channel), but how can you configure it from the outside?

No way.

For this, each voice has an independent parametric MIDI channel. To avoid problems, tracks should have different values for pCH.

If you want to operate in classic mode, simply set all channels of track X to the same MIDI channel.



INTF PAGE

Interface Settings Group.

BPM BPM PARAMETER

VALUE FROM: 1i:255i (MIDI: 1i:127i) SCALE: RISING

Sets the tempo for the track.

Works as a global clock for all xPER, RISE and FALL time parameters has a wide range of values from 1 to 255. Yes, that's right — you can set one beat per minute.

TSEL TSEL PARAMETER (TRACK SELECTOR TYPE)

VALUE FROM: 1i:4i SCALE: ONE-HOT

Controls the track selector sensor mode.

- HOLD allows you to activate/deactivate envelopes, as well as select a track to change parameters with the encoder.
 - DRONE allows you to activate envelopes, as well as select a track to change parameters with the encoder. Deactivation of the envelope occurs by pressing the track selector again or by MIDI noteOn message.
- For more details, see [\[TRACK SELECTOR MODES\]](#).

EDIT — allows only selecting tracks to change encoder parameters. Envelopes can only be activated via MIDI in this mode

WSEL WSEL PARAMETER (WAVEFORM SELECT)

VALUE FROM: 1i:16i SCALE: BINARY

Controls the selection of tracks whose waveforms will be displayed on the oscilloscope.

WSYN WSYN PARAMETER (WAVEFORM SYNC)

VALUE FROM: 1i:4i SCALE: ONE HOT

Controls the selection of the track to which the waveform on the oscilloscope will be synchronized.



PRST PAGE

This page is used to save and load presets.

A preset is a set of track parameters.

Presets should be perceived as a mechanism for copying information from one track to another.

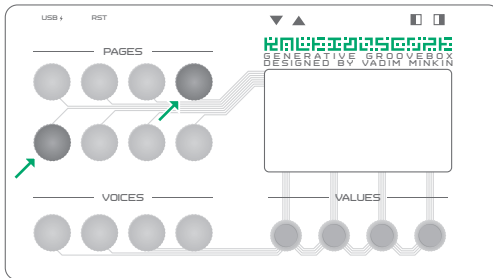
SLOT: SLOT PARAMETER (SLOT)

VALUE FROM: 0i:127i SCALE: NUMERIC

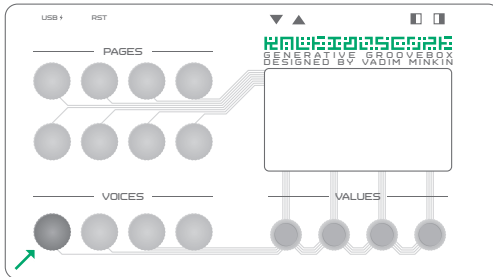
Selecting a slot for loading a preset from memory into a track.

Select a slot to save a preset from a track to memory.

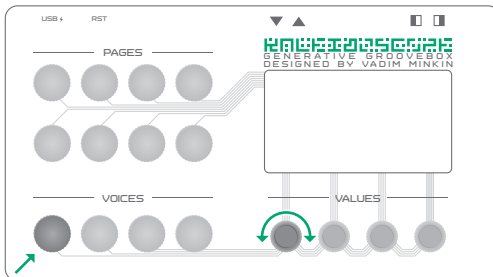
Go to the CFG-PRST page.



Hold down the desired Tx track selector.

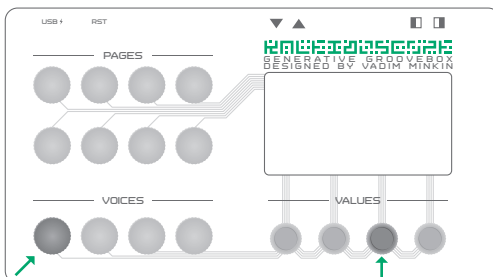


Turn the V1 encoder to select a slot.



Hold down the desired Tx track selector.

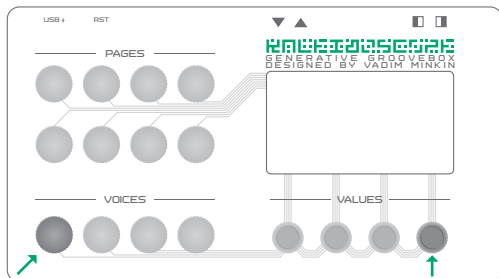
Hold down the V3 encoder to load the contents of the selected slot of this track into the current preset.



Keep the encoder pressed until the growing scale on the screen is full.

! This action will permanently erase the current audio, unless you have previously saved it to one of the slots or as a file on your computer.

Hold down the desired Tx track selector.



Hold down the V4 encoder to save the contents of the current preset into the selected slot.

! This action will permanently erase the preset stored in the selected slot, unless you previously saved it to another slot or as a file on your computer.

Each of the 128 presets (0:127) has an editable name.

To edit the name, use encoder2.

1. Hold down the track selector to select the track whose slot name you want to edit.
2. Turning encoder2 selects the letter at the current cursor position
3. Turning the pressed encoder2 selects the cursor position in the line

The name is saved automatically to memory. While the current name is not saved or is being read from memory, the # sign lights up at the end of the name line. When transferring a preset to another device, the name is also transferred.



SCOP PAGE

This page displays the waveform and spectrum of the sum of the tracks selected by the wSEL parameter (INTF page).

PART 3. GENERATIVE SEQUENCER

KALEIDOSCOPE'S generative sequencer does not exist as a separate module. It is rather a philosophy that relies on several simple features embedded in the synthesizer's control system. Using these features, you create sequences and patterns.

KALEIDOSCOPE'S generative sequencer does not conflict with an external sequencer or keyboard, since it is an extension of the sequencer's functions, and works in the area of parameter modulation, not their direct control.

The generative sequencer is a very advanced modular arpeggiator that can cycle through not only the frequency values of voices, but also any other parameters. Including those that also have musical meaning and are of an octave nature.

ENVELOPE FEATURES

■ TRIGGER EXCHANGE

To run a volume envelope or a modulation envelope, you need to send it a trigger.

You can generate a trigger by touching the track selector (built-in keyboard) or by sending a noteOn.

MIDI message from the channel of this envelope (external keyboard) but KALEIDOSCOPE has another way — generating an EOC (end of cycle) trigger.

Each envelope generates an EOC trigger at the end of the attack-decay cycle. This trigger can be sent to any of the envelopes of the same type.

- A volume envelope can send a trigger to any combination of volume envelopes.
- A modulation envelope can send a trigger to any combination of modulation envelopes.

Creating sequences on Kaleidoscope consists of skillfully patching EOC triggers between envelopes.

To enable the transmission of the EOC trigger for a track, you need to set the EOC parameter on the vSEQ or mSEQ pages to on.

The EOC trigger is sent only when the envelope is held down either from the track selector or via MIDI.

To select which envelopes will transmit the trigger, you need to set the LNK parameter on the same page.

▣ PATTERN MASK

Let's say the volume envelope vca1 is set to send the EOC trigger only to itself. In this case, it will retrigger itself as long as the volume envelope 1 is held down.

To avoid holding down a sensor or a key on the keyboard, set the TSEL parameter of track 1 to DRONE (value 1). Now it is enough to activate the envelope once and it will be held until the next double-click on the track selector or MIDI noteOn message (noteOn must contain the same note as when the envelope was triggered).

Each envelope has a pattern mask with adjustable length.

The PATN parameter controls whether the envelope will have an effect at this step of the retrigger or not.

If the pattern at the current step is 1, then we will hear the effect of the envelope. Otherwise, the envelope will continue to work in shadow mode, but we will not hear its effect.

Each reception of the EOC trigger increases the pattern cursor by one. If the cursor value exceeds the mask length, its value is reset to 0.

Activating the envelope via the track selector or MIDI noteOn message always resets the pattern cursor to the first step.

If the transmission of the EOC trigger is disabled, the PATN parameter value is ignored — the envelope will be in effect.

The pattern and its length are just parameters, i.e. ordinary numbers that can be modulated.

▣ RHYTHMIC NATURE

The PERI parameter values have a rhythmic nature:

- The durations of the periods are related to the durations of the notes (eighth, sixteenth, triplet, etc.), which makes it easy to generate patterns with a large but limited period.
- The durations of the periods double every 12 encoder clicks (or 1 click of the pressed encoder) — just like the note frequency.

The PERI parameter values are also related to the set tempo. Once you create a pattern or some timbral dynamics, you will not lose it by changing the tempo.

When changing the PERI parameter of the envelope, the value in the range from 64 to 0.041 is displayed on the screen.

The higher the value, the higher the speed of the period. But in reality, the meaning of this value is deeper. If you take the inverse PERI value, the attack-decay period will last exactly this part of the measure at the current BPM.

For example.

With $PERI = 8$, the period length is equal to $1/8$ of a bar at the current BPM.

With $PERI = 0.125$, the period length is equal to $1/0.125$, i.e. 8 bars at the current BPM.

With $PERI = 7$, the envelope period length is equal to $1/7$ of a bar at the current BPM.

This approach to control allows you to perfectly synchronize the envelopes with each other.

For example, an envelope with a period of $8 (1/8)$ of a bar and a pattern length of 8 steps will have a full pattern period equal to 1 bar.

An envelope with a period of $7 (1/7)$ of a bar and a pattern length of 7 steps will have a full pattern period equal to 1 bar.

Thus, having two envelopes with different periods, patterns and shapes, we still get a full period equal to one bar.



The 0:35 comb filter values, which are tied to the current BPM of the track, work similarly.

■ MODULATOR QUANTIZATION

Each modulation envelope has QNT1 (Quantize) and QNT2 parameters, which allow you to make the triangular envelope shape stepped.

The QNT parameter allows you to turn the signal into a step signal with a controlled step size:

- 0 no quantization
- 1 quantization by semitones
- 2 quantization by tone
-
- 4 quantization by major thirds
- 5 quantization by fourths
-
- 7 quantization by fifths
- ...
- 12 quantization by octaves
- ... and so on.

For example, using octave quantization, you can easily create an arpeggiator. It is enough to apply the action of the modulator to the frequency of the track. In this case, the rhythm of the arpeggiator is not connected with the rhythm of the main voice, which allows you to create a lot of new effects, and even modulate the rhythmic characteristics of the modulator itself.

Let's not forget that there are other parameters that can also be "octave" modulated. For example, you can create an arpeggiator that will change the filter resonance frequency or the delay length of the combo filter.

Let me remind you that the duration of the PERI parameter of all envelopes is also subject to octave doubling. This allows you to double the rhythm speed using a quantized modulator.

You can add up the values of modulators with different quantizations. For example, if you add up modulators with octave quantization and fifth quantization. The resulting signal will highlight only the values that are octaves, fifths away from the original value. Also, this total modulating signal can have a complex shape that is modulated in time by some other modulators.

The QNT parameter can also be modulated.

NOTE QUANTIZATION

Each track has a SCAL (Scale) parameter, which allows you to quantize the oscillator frequency signal by some known scales:

OFF — quantization is disabled

nMAJ — natural major

nMIN — natural minor

PENT — major pentatonic

HEXA — hexatonic

BLUE — blues hexatonic

EMP — empty scale. Will be filled in future firmware

FIX — the oscillator ignores the value of MIDI notes. The frequency is adjusted only by the TRAN (transposition) parameter

For example, let's say natural major is selected (SCAL = nMAJ). Zero transposition (TRAN = 0), this means the key of C is selected. If you modulate the frequency with a linear signal from the modulator, you will hear a simple selection of notes C, D, E, F, G, A, B, Since other notes are not allowed in this key.

If you make the transposition equal to 2 (TRAN = 2), then you will choose the key of D and so on.

If you apply quantization to a modulator by some integer, then the modulator signal will become linear. And as a result, it will select only some notes from the scale.

If you add the signals of two different modulators, then the sequence of notes will be more complex. It will depend on the parameters of the modulators — their shape, period, quantization and modulation depth.

■ ■ COMBINING WITH AN EXTERNAL SEQUENCER OR KEYBOARD

KALEIDOSCOPE'S generative sequencer works as an add-on to the envelopes and an extension of the modulation capabilities. It does not conflict with an external keyboard or sequencer, but allows you to complicate and diversify the methods of controlling the synthesizer and its timbres.

■ ■ EXAMPLES

By combining Trigger Exchange, Pattern Mask, Rhythmic Nature and Modulator Quantization you get a way to generate a huge number of dynamic rhythmic patterns not only at the envelope level, but also at the modulation level and at the level of the sequence itself, which can be modulated.

1. Cyclic envelope
2. Add a cyclic modulation envelope that changes some timbre parameter.
3. Add a pattern to both envelopes.
4. Add a period modulation of the volume envelope with octave quantization so that the frequency doubles
5. Add a pattern modulation of the volume envelope modulator.
6. Add echo
7. Enjoy!

MISCELLANEOUS

MANAGING PRESETS

You can save the current track settings to one of 128 memory slots.
You can load the settings of one of 128 memory slots into a track.
You can export slots with track settings from KALEIDOSCOPE to your computer.
You can import slots with track settings from your computer into KALEIDOSCOPE.

!! SAVING AND LOADING

Each track has 4 native slots for storing settings.
Settings from these tracks can be saved to a memory slot.
Settings from a memory slot can be loaded into a native track slot.
Settings from one slot cannot be directly loaded into another slot.

Go to the CFG-PRST page.
Hold down the desired track selector sensor Tx.
Turn the V1 encoder to select the slot.

Hold down the desired Tx track selector.
Hold down the V3 encoder to load the contents of the selected slot of this track into the current preset.

Keep the encoder pressed until the on-screen progress bar is full.



This action will permanently erase the current native preset unless you have previously saved it to one of the slots or as a file on your computer.

Hold down the desired Tx track selector.
Hold down the V4 encoder to save the contents of the current preset into the selected slot.

Keep the encoder pressed until the on-screen progress bar is full.



This action will permanently erase the preset stored in the selected slot unless you have previously saved it to another slot or as a file on your computer.

EXPORTING AND IMPORTING PRESETS

To export a preset from KALEIDOSCOPE to the outside world, you will need a computer.

1. Connect the device to the computer via USB.
2. Hold down the encoder button 1.
3. Press the reset button.
4. Release the encoder button 1.
5. The computer should detect the device as a flash drive. A message with instructions on updating the firmware will appear on the KALEIDOSCOPE screen.
6. Copy a file called exp.bin to the root of the flash drive.
7. Copy a file (or several) called expXXX.bin to the root of the flash drive, where XXX is the preset number.
 - It is important that the number has 3 digits. For example, to export slot 3, the name will be exp003.bin. To export slot 29, the name will be exp029.bin.
8. Safely remove the device from the computer's file system.
9. Press the reset button to restart KALEIDOSCOPE.
10. The application will detect the export request, as there is a file export.bin there
11. The application will put the settings from the corresponding XXX slots into the files expXXX.bin.
12. After that, the device will delete the file export.bin and start up in USB mode again. You can save the exported presets on your computer.

To import a preset from the outside world into KALEIDOSCOPE, you will need a computer.

1. Connect the device to the computer via USB.
2. Hold down the encoder button 1.
3. Quickly press the reset button.
4. Release the encoder button 1.
5. The computer should detect the device as a flash drive. A message with instructions on updating the firmware will appear on the Kaleidoscope screen.
6. Copy the file named imp.bin to the root of the flash drive.
7. Rename the presets you want to import into KALEIDOSCOPE in advance in the impXXX.bin format.
 - It is important that the number has 3 digits. For example, to import a preset into slot 5, the name should be imp005.bin. To import a preset into slot 33, the name should be imp033.bin.
8. Copy a file (or several) named impXXX.bin to the root of the flash drive, where XXX is the number of the slot where you want to import the settings.
9. Safely remove the device from the computer's file system.
10. Press the reset button to restart KALEIDOSCOPE.
11. The application will detect the import request, since there is a file import.bin there
12. The application will place the settings from the impXXX.bin files in the corresponding XXX slots
13. After that, the device will delete the import.bin file and proceed to executing the application.

FIRMWARE UPDATE

1. Connect the device to the computer via USB.
2. Hold down the encoder button 1.
3. Press the reset button.
4. Release the encoder button 1.
5. The computer should detect the device as a flash drive. A message with instructions on updating the firmware will be displayed on the KALEIDOSCOPE screen.
6. Rename the firmware file to blink.bin in advance.
7. Copy the firmware file to the flash drive.
8. Safely remove the device from the computer's file system.
9. Press the reset button to restart KALEIDOSCOPE.
10. The firmware will be copied from the file to the device memory.
11. The firmware file will be deleted.
12. The device will go to the application.

Simplified Font

The kaleidoscope interface can be launched with a simplified font.

To do this, you need to hold down encoder2 while the device is starting up.

1. Press and hold encoder2
2. Press the RST button to restart the firmware
3. When the interface loads, you should see the font change
4. You can release encoder2

VALUE SCALES

On the KALEIDOSCOPE display, values are mapped to rough scales.

There are several types of scales.



It seems like magic, but by the location and type of scales on the page, our brain begins to understand which page of KALEIDOSCOPE you are on and which parameter you need to adjust.

The minimum value from a MIDI point of view is always located at the bottom of the scale — 0.

The scale shows what happens to the parameter as this value increases.

For example, the frequency of an LP filter with resonance (FLT parameter) has an rising scale.

As the value increases, the bar of the scale takes up more and more space. It also shows the frequencies that the filter passes.

The delay length (DELA) has a falling scale. The delay lengths are set according to the periods of midi notes. Accordingly, the shortest delay (the highest frequency) is at the top, and the longest delay (the lowest frequency) is at the bottom.

The symmetry parameter of the triangle oscillator MID has a bidirectional scale. In the middle, the parameter value is 0.5, and the oscillator shape is a symmetrical triangle. When the parameter is shifted relative to 0.5 towards 0 or 1, the symmetry is broken, and the number of harmonics increases. Up to the maximum value for the saw in values 0 and 1. Similarly, the SHP parameter for envelopes has a bidirectional scale.

The FOLD, MIX, DRIV, FBK, VOLU parameters have a bidirectional scale. All these parameters have two operating modes depending on the parameter value. If the parameter is greater than 0, then this is one operating mode, if less than 0, then another. Moreover, its intensity increases as it moves away from zero.

Similarly, all DEP modulation depth parameters have a bidirectional scale. Since the modulation effect can be both positive and negative.

Some parameters are responsible for selecting one option from several. For this, various one-hot scales are used.

Selecting MIDI channels vCH, mCH, pCH is always a choice of one from several.

Selecting a parameter for PAR modulation — choosing 1 from 32.

Selecting a track for DST modulation — choosing 1 from 4.

EOC envelope cyclic mode on/off — choosing 1 from 2.

Some parameters are responsible for selecting a combination. For this, a binary scale is used.

This scale shows which of the 4 tracks are involved in the interaction.

LINK parameter for selecting EOC trigger recipients — choosing 1 from 16 combinations.

FMSR modulation source selection — 1 of 16 combinations.

SEND audio output voice sum selection — 1 of 16 combinations.

PATN envelope pattern — 1 of 256 combinations.

POLYPHONY

■ POLYPHONY AND MIDI CHANNELS

Each volume envelope and modulation envelope has its own MIDI channel. noteOn/noteOff messages received from these channels start/release the envelopes.

The channel of volume envelope 1 does not have to be the same as the channel of modulation envelope 1.

You can assign them to different channels to start them independently.

If you assign several envelopes to one channel, then receiving a noteOn/noteOff message from this channel will start/release all these envelopes at once.

The volume envelope can be set to a MIDI channel in monophonic or polyphonic mode. The choice of mode is determined by the prefix before the channel number on the display: mNN for monophonic and pNN for polyphonic.

If several volume envelopes and modulation envelopes are set to the same channel NN in monophonic mode, then receiving a note from this channel will start all these channels.

If several volume envelopes are set to the same NN channel in polyphonic mode. Then receiving a note from this channel will trigger a free volume envelope. If all voices in the polyphonic channel are occupied, the note will replace the oldest note on these voices.

Polyphonic note distribution does not apply to modulation envelopes.

In addition to selecting one of the 16 midi channels, the modulation envelope has an additional feature — automatic triggering together with one of the volume envelopes.

This mode is useful when you want to trigger the modulation envelope together with one of the volume envelopes that works in polyphonic mode.

■ ACTIVATION AND HOLD INDICATOR

In the lower left corner of the KALEIDOSCOPE screen there is an important indicator of track activation and hold.

The upper row of the indicator shows which tracks are **active**, i.e. turning the encoder will change their parameters.

The lower row of the indicator shows which tracks are **held**, i.e. their envelopes are working, and we hear the sound or the modulation action..

■ HOLD, DRONE AND EDIT

The track selector can work in three modes — HOLD, DRONE and EDIT. The CFG-INTF-TSEL parameter is responsible for their selection.

Touching the track sensor and then holding it in any mode **activates** the track for subsequent editing of its parameters with the encoder.

Releasing the track sensor in any mode **deactivates** the track for editing with the encoder.

The track is in HOLD mode (CFG-INTF-TSEL = 0):

- touching the track sensor and then holding it
 - starts and holds the track volume envelope, as well as the modulation envelope,
- releasing the sensor
 - releases the track volume envelope and modulation envelope.

The track is in DRONE mode (CFG-INTF-TSEL = 1):

- quick double-click on the sensor:
 - starts the track volume envelope and modulation envelope, if they were previously released.
 - releases the track volume envelope and modulation envelope, if they were previously launched.

Track works in EDIT mode (CFG-INTF-TSEL = 2):

- does not start or releases envelopes
 - you can start an envelope in this mode only via MIDI

■ EFFECT OF MIDI NOTES

KALEIDOSCOPE is intuitive.

You select a voice on the track selector and turn the encoder to change the parameters.

But if you pressed a note on the MIDI keyboard and are already holding the voice, do you need to press the track selector to change the parameter?

Of course you don't!

You press a note on the keyboard and turn the encoder to change the parameter of the selected track.

MIDI noteOn holds the track (visible on the hold indicator) and activates the track (visible on the activation indicator).

But what if your external sequencer plays on several KALEIDOSCOPE tracks, that is, holds and activates several tracks at the same time. And you want to change the parameter of one without having to stop the sequencer. How can you do this?

To do this, you need to press the track selector sensor. At this point, KALEIDOSCOPE will ignore MIDI activation via noteOn and will only consider pressing the track selector.

It sounds quite complicated, but this is what you expect from a device for a seamless user experience.

Your hands will automatically reach for the track selector while the sequencer plays the beat.

MIDI PARAMETER TABLE

PARAMETER	SCALE	CC	DESCRIPTION
ADTV ▪ ADDITIVE PARAMETERS			
MIDI	Bidir	10	Triangle Osc Asymmetry
FOLD	Bidir	11	Fold level
MIX	Bidir	12	Mix
DRIV	Bidir	7	Drive
SBTV ▪ SUBTRACTIVE PARAMETERS			
FILT	Rising	17	Cut off frequency
RESO	Rising	18	Resonance
DELY	Falling	32	Delay
FEDB	Bidir	13	Feedback
FREQ ▪ OSCILLATOR FREQUENCY PARAMETER			
TRAN	Bidir	23	Transpose
FINE	Bidir	6	Fine tuning
GLID	Rising	15	Glide
SCALE	One-Hot	39	Incoming note Quantize
MAST ▪ MASTER SECTION			
VOLU	Bidir	16	Volume
FMVO	Bidir	34	FM signal Amplitude
FMSR	Binary	14	FM source
SEND	Binary	84	Audio out configuration
vENV ▪ VOLUME ENVELOPE SHAPE			
SHAP	Bidir	9	Attack-Decay asymmetry
PERI	Rising	8	Attack-Decay period
SUST	Rising	19	Sustain
RELE	Rising	35	Release
PARAMETER	SCALE	CC	DESCRIPTION
vSEQ ▪ VOLUME ENVELOPE SEQUENCER			
EOC	One-Hot	22	EOC trigger on/off
LINK	Binary	21	EOC link
PATN	Binary	5	Pattern-mask
PSIZ	One-Hot	4	Pattern-mask lenght
vMX1 ▪ VOLUME ENVELOPE MODULATION 1			
PARM	One-Hot	49	Parameter
DEST	One-Hot	50	Destination track
DEPT	Bidir	48	Modulation depth
ALIA	~	51	Alias

vMX2 ▪ VOLUME ENVELOPE MODULATION 2

PARM	One-Hot	53	Parameter
DEST	One-Hot	54	Destination track
DEPT	Bidir	52	Modulation depth
ALIA	~	55	Alias

mENV ▪ MODULATION ENVELOPE SHAPE

SHAP	Bidir	24	Attack-Decay asymmetry
PERI	Rising	25	Attack-Decay period
QNT1	One-Hot	33	Quantize 1
QNT2	One-Hot	20	Quantize 2

mSEQ ▪ MODULATION ENVELOPE SEQUENCER

EOC	One-Hot	27	EOC trigger on/off
LINK	Binary	28	EOC link
PATN	Binary	56	Pattern-mask
PSIZ	One-Hot	57	Pattern-mask lenght

mMX1 ▪ MODULATION ENVELOPE MODULATION 1

PARM	One-Hot	30	Parameter
DEST	One-Hot	31	Destination track
DEPT	Bidir	29	Modulation depth
ALIA	~	37	Alias

mMX2 ▪ MODULATION ENVELOPE MODULATION 2

PARM	One-Hot	45	Parameter
DEST	One-Hot	46	Destination track
DEPT	Bidir	44	Modulation depth
ALIA	~	47	Alias

PARAMETER	SCALE	CC	DESCRIPTION
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MIDI ▪ MIDI CHANNELS

vCH	One-Hot	43	Vol Envelope MIDI channel
mCH	One-Hot	42	Mod Envelope MIDI channel
pCH	One-Hot	116 + i	Parameter MIDI channel

INTF ▪ INTERFACE

BPM	Rising	38	Tempo
TSEL	One-Hot	36	Track Selector Mode
WSEL	Binary	40	Waveform select
WSYN	One-Hot	41	Waveform sync

PRST ▪ PRESET

SLOT	One-Hot	114	Preset slot
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aENV ▪ AFTERTOUCH ENVELOPE SHAPE

RISE	Rising	61	Rising speed
FALL	Rising	62	Falling speed

aMX ▪ AFTERTOUCH ENVELOPE MODULATION

PARM	One-Hot	58	Parameter
DEST	One-Hot	59	Destination track
DEPT	Bidir	60	Modulation depth
ALIA	~	63	Alias

fENV ▪ VOLUME ENVELOPE SHAPE

RISE	Rising	67	Rising speed
FALL	Rising	68	Falling speed

fMX ▪ VELOCITY ENVELOPE MODULATION

PARM	One-Hot	64	Parameter
DEST	One-Hot	65	Destination track
DEPT	Bidir	66	Modulation depth
ALIA	~	69	Alias

kENV ▪ KEYTRACK ENVELOPE SHAPE

RISE	Rising	73	Rising speed
FALL	Rising	74	Falling speed

kMX ▪ KEYTRACK ENVELOPE MODULATION

PARM	One-Hot	70	Parameter
DEST	One-Hot	71	Destination track
DEPT	Bidir	72	Modulation depth
ALIA	~	75	Alias

PARAMETER	SCALE	CC	DESCRIPTION
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wMX ▪ MODWHEEL MODULATION

PARM	One-Hot	76	Parameter
DEST	One-Hot	77	Destination track
DEPT	Bidir	78	Modulation depth
ALIA	~	79	Alias

pMX ▪ PITCH BEND MODULATION

PARM	One-Hot	80	Parameter
DEST	One-Hot	81	Destination track
DEPT	Bidir	82	Modulation depth
ALIA	~	83	Alias

CREDITS

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